

The Second Scaling Law: Intelligence Scales with Parameters, Wisdom Scales with Architecture

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Abstract

Neural scaling laws [12, 11] established that model intelligence—measured by loss on held-out data—scales predictably with parameter count and training data. We demonstrate that a second, orthogonal scaling law governs *wisdom*: the capacity to learn from sequential experience and transfer insights to novel situations. Through formal analysis and empirical validation on 1,200 agent–task–round interactions across three frontier LLMs, we prove that intelligence and wisdom are asymptotically decorrelated (Theorem 1), that wisdom is dominated by architectural properties rather than parameter count (Theorem 2), and we derive a functional form for the *Wisdom Scaling Law* governed by three measurable architectural quantities: *plasticity*, *immunity*, and *homeostasis* (Theorem 3). Our framework unifies recent advances in liquid neural networks, cognitive immunity, and Bayesian self-regulation under a single theoretical umbrella, and yields falsifiable predictions distinguishing it from the established parameter-scaling paradigm. We contend that the first scaling law charts the ceiling of artificial intelligence; the second charts the floor of artificial wisdom.

1 Introduction

The discovery of neural scaling laws [12] transformed deep learning from an empirical craft into a predictive science. The central finding—that cross-entropy loss decreases as a power law of parameter count N , dataset size D , and compute budget C —provided the theoretical bedrock for the scaling race from GPT-2 (1.5B) to GPT-4 (estimated $>1T$) and beyond. The Chinchilla revision [11] refined the optimal allocation between N and D , but preserved the core thesis: *more is better*.

This paper identifies a fundamental blind spot in the scaling paradigm.

The anomaly. Consider an LLM agent that produces a flawed safety response. When re-presented with a structurally identical scenario, state-of-the-art models—regardless of their parameter count—fail at rates statistically indistinguishable from their first attempt [14, 17]. A 1-trillion-parameter model repeats the same class of error as a 7-billion-parameter model. Scaling parameters improved *intelligence* (first-attempt score) but did not improve *wisdom* (the ability to learn from and not repeat mistakes).

This observation is not an isolated curiosity. Our empirical study across three frontier LLMs (DeepSeek-v4-flash, Qwen-Plus, Qwen-Max), four memory strategies, and 20 tasks evaluated over 5 rounds with 3 seeds each (Section 5, $N=3,600$ evaluations) reveals a *negative* correlation between intelligence and wisdom: Spearman $\rho(I, W) = -0.389$ ($p = 0.212$, $n=12$)—a medium effect by Cohen’s convention. Qwen-Plus achieves 60% higher intelligence ($I=2.84$) but *half* the wisdom ($W=0.071$), a *ceiling effect* in which high initial performance eliminates the headroom required for learning.

Our contribution. We propose that wisdom is governed by a *second scaling law*, orthogonal to the first:

$$\text{First Law (Kaplan): } I \propto N^a \cdot D^b \quad (1)$$

$$\text{Second Law (Ours): } W \propto \Phi(\mathcal{A})^\alpha \cdot \Psi(\mathcal{A})^\beta \cdot H(\mathcal{A})^\gamma \cdot |\mathcal{E}|^\delta \quad (2)$$

where Φ , Ψ , H are architectural properties (*plasticity*, *immunity*, *homeostasis*) formally defined in Section 3, and $|\mathcal{E}|$ is the number of interactive episodes.

The key structural difference: the first law’s dominant terms (N , D) are *resource quantities* that can be increased by spending more money. The second law’s dominant terms (Φ , Ψ , H) are *architectural qualities* that require design

innovation. This distinction has profound implications for the field: the path to wiser AI is not paved with more GPUs, but with better cognitive architectures.

We make three formal contributions:

1. **Theorem 1** (Intelligence–Wisdom Decorrelation): For any fixed architecture, intelligence and wisdom are asymptotically decorrelated as parameter count grows.
2. **Theorem 2** (Architectural Dominance): A more plastic architecture with fewer parameters can match or exceed the wisdom of a less plastic architecture with arbitrarily more parameters.
3. **Theorem 3** (Wisdom Scaling Law): Under three axioms, expected wisdom follows the power-law form in Eq. 2.

2 Preliminaries and Formal Setup

2.1 Agent Model

We formalize an AI agent as a tuple $M = (\theta, \mathcal{A})$ where $\theta \in \mathbb{R}^N$ denotes the parameter vector ($N = |\theta|$) and $\mathcal{A}$ denotes the *cognitive architecture*—the structural specification of how the agent processes, stores, and retrieves experiential information during sequential interactions.

Crucially, $\mathcal{A}$ encompasses not only the neural network topology but also the memory system, feedback integration mechanism, and self-regulation strategy. Two agents with identical parameters but different architectures (e.g., a bare LLM vs. the same LLM augmented with cognitive immunity) are formally distinct: $M_1 = (\theta, \mathcal{A}_1) \neq M_2 = (\theta, \mathcal{A}_2)$.

2.2 Task and Evaluation Protocol

Let $\mathcal{T}$ be a distribution over tasks. Each task $t \sim \mathcal{T}$ is presented to agent M for R sequential *rounds*. Between rounds, the agent receives a feedback signal f_r (the score and optionally an explanation of the failure mode). We denote:

- $s_r(M, t) \in [0, S]$: the score of agent M on task t in round r
- $\bar{s}_1(M) = \mathbb{E}_{t \sim \mathcal{T}}[s_1(M, t)]$: the expected first-round score (*intelligence*)
- $\bar{s}_R(M) = \mathbb{E}_{t \sim \mathcal{T}}[s_R(M, t)]$: the expected final-round score

Definition 1 (Intelligence). *The intelligence of agent M is its expected performance on first encounter:*

$$I(M) = \mathbb{E}_{t \sim \mathcal{T}}[s_1(M, t)] \quad (3)$$

Definition 2 (Wisdom Quotient). *The Wisdom Quotient of agent M is the expected relative improvement from first to final round:*

$$W(M) = \mathbb{E}_{t \sim \mathcal{T}} \left[\frac{s_R(M, t) - s_1(M, t)}{S} \right] \quad (4)$$

where S is the maximum possible score, ensuring $W \in [-1, 1]$.

Remark 1. $W > 0$ indicates net learning; $W < 0$ indicates net regression (the agent gets worse with experience); $W = 0$ indicates no learning. Importantly, if $s_1 = S$ (already perfect), then $W = 0$ regardless of architecture—the agent has hit the intelligence ceiling and wisdom becomes unmeasurable for that task. This ceiling effect is informative: it reveals tasks where wisdom is structurally irrelevant.

2.3 The Intelligence–Wisdom Plane

Every agent M maps to a point $(I(M), W(M))$ in the *IW-plane*. The first scaling law predicts that increasing N moves M rightward along the I -axis. Our central question is: *what moves M upward along the W -axis?*

3 Three Architectural Axioms

We now formalize three properties of cognitive architectures that we hypothesize to be the dominant factors governing wisdom. Each axiom defines a measurable quantity and states a minimal structural requirement.

Axiom 1 (Plasticity Φ). *An architecture $\mathcal{A}$ has plasticity $\Phi(\mathcal{A}) \geq 0$ if it can modify its effective computational mapping $g_{\mathcal{A}} : \mathcal{X} \rightarrow \mathcal{Y}$ in response to runtime experience $e \in \mathcal{E}$ without modifying the base parameters θ . Formally:*

$$\Phi(\mathcal{A}) = \mathbb{E}_{e \sim \mathcal{E}} [\|g_{\mathcal{A}}(\cdot | e) - g_{\mathcal{A}}(\cdot | \emptyset)\|_{\mathcal{F}}] \quad (5)$$

where $\|\cdot\|_{\mathcal{F}}$ is a function-space norm (e.g., L^2 under the task distribution) and $\emptyset$ denotes the null experience.

Remark 2. $\Phi = 0$ for a frozen transformer with no memory: the same input always produces the same output distribution regardless of prior experience. $\Phi > 0$ for architectures with runtime-modifiable components: memory-augmented networks, liquid neural networks [10], retrieval-augmented generation, and the cognitive immunity system [1].

Axiom 2 (Immunity Ψ). *An architecture $\mathcal{A}$ has immunity $\Psi(\mathcal{A}) \in [0, 1]$ measuring its capacity to extract transferable failure rules. Given a failure event f on task t_1 and a structurally related but distinct task t_2 :*

$$\Psi(\mathcal{A}) = \mathbb{E}_{f, t_1, t_2} [P(s_1(M, t_2) > s_{base} \mid \text{observed } f \text{ on } t_1, t_2 \sim \mathcal{T}_{t_1})] - P_{base} \quad (6)$$

where $\mathcal{T}_{t_1}$ is the distribution of tasks structurally related to t_1 , and P_{base} is the baseline success probability without failure information.

Remark 3. Ψ measures transfer: not whether the agent avoids repeating the exact same mistake (memorization), but whether it extracts a principle applicable to novel situations. A lookup table has $\Psi = 0$ (perfect recall, zero transfer). The proposed cognitive immunity framework is designed to maximize Ψ through antigen extraction and antibody generalization [1].

Axiom 3 (Homeostasis H). *An architecture $\mathcal{A}$ has homeostasis $H(\mathcal{A}) \in [0, 1]$ measuring its capacity to maintain stable performance under distributional shift without external recalibration:*

$$H(\mathcal{A}) = 1 - \frac{\text{Var}_{t \sim \mathcal{T}'} [W_t(M)]}{\text{Var}_{t \sim \mathcal{T}} [W_t(M)]} \quad (7)$$

where $\mathcal{T}$ is the training distribution and $\mathcal{T}'$ is a shifted distribution. $H = 1$ indicates perfect robustness; $H = 0$ indicates that distributional shift fully destabilizes wisdom.

Remark 4. Homeostasis captures the stability requirement: an agent that achieves high W on one task distribution but collapses on another has low H . Bayesian self-regulation [18] provides one mechanism for achieving high H through posterior-guided parameter adaptation; PID-style control [3] provides another through error-driven feedback.

3.1 Relationship Between Axioms

The three axioms are designed to be *orthogonal*: Plasticity concerns the capacity for change; Immunity concerns the quality of change (transferability); Homeostasis concerns the stability of change. An architecture can have high Φ but low Ψ (it changes a lot but learns nothing transferable), or high Ψ but low H (it transfers well but is brittle to distributional shift). Wisdom, we argue, requires all three.

4 Main Theorems

Theorem 1 (Intelligence–Wisdom Decorrelation). *Let $\{M_n\}_{n \in \mathbb{N}}$ be a family of agents with fixed architecture $\mathcal{A}$ and increasing parameter count $n = |\theta|$. Assume:*

- (i) $I(M_n) \rightarrow I^*(\mathcal{A})$ as $n \rightarrow \infty$ (intelligence saturates);
- (ii) The architecture $\mathcal{A}$ has finite memory capacity $C(\mathcal{A}) < \infty$.

Then $W(M_n) \rightarrow W^*(\mathcal{A})$ where W^* depends only on $\mathcal{A}$ and not on n . Consequently:

$$\lim_{n \rightarrow \infty} \frac{\partial W(M_n)}{\partial n} = 0 \quad (8)$$

That is, beyond a sufficient parameter count, adding more parameters yields zero marginal wisdom.

Proof. By Definition 2, $W(M) = \mathbb{E}_t[(s_R - s_1)/S]$. The improvement $s_R - s_1$ is determined by the agent's ability to integrate feedback from rounds $1, \dots, R-1$.

Under assumption (ii), the architecture $\mathcal{A}$ can store at most $C(\mathcal{A})$ bits of experiential information between rounds. The wisdom-relevant computation at round $r > 1$ is:

$$s_r(M_n, t) = h_{\mathcal{A}}(\theta_n, t, \text{mem}_r(\mathcal{A}))$$

where $\text{mem}_r(\mathcal{A})$ is the memory state, bounded by $|\text{mem}_r| \leq C(\mathcal{A})$.

As $n \rightarrow \infty$, $h_{\mathcal{A}}(\theta_n, t, m) \rightarrow h_{\mathcal{A}}^*(t, m)$ for each fixed m (by universal approximation and assumption (i)). Since $\text{mem}_r(\mathcal{A})$ is determined solely by $\mathcal{A}$ and the feedback sequence $f_1, \dots, f_{r-1}$ (independent of n for sufficiently large n), we have:

$$s_r(M_n, t) \rightarrow h_{\mathcal{A}}^*(t, \text{mem}_r(\mathcal{A}))$$

which is independent of n . Therefore $W(M_n) \rightarrow W^*(\mathcal{A}) = \mathbb{E}_t[(h_{\mathcal{A}}^*(t, \text{mem}_R) - h_{\mathcal{A}}^*(t, \emptyset))/S]$, which depends on $\mathcal{A}$ alone. $\square$

Theorem 2 (Architectural Dominance). *Let $\mathcal{A}_1, \mathcal{A}_2$ be two architectures satisfying $\Phi(\mathcal{A}_1) > \Phi(\mathcal{A}_2)$ and $\Psi(\mathcal{A}_1) \geq \Psi(\mathcal{A}_2)$. Then for any $\epsilon > 0$, there exists a threshold N^* such that:*

$$W(M_{\mathcal{A}_1, n_1}) \geq W(M_{\mathcal{A}_2, n_2}) - \epsilon \quad \forall n_1 \geq N^*, \forall n_2 \in \mathbb{N}$$

That is, a sufficiently large model with the superior architecture achieves wisdom within ϵ of any model with the inferior architecture, regardless of how many parameters the inferior architecture has.

Proof. By Theorem 1, for each architecture $\mathcal{A}_i$:

$$W(M_{\mathcal{A}_i, n}) \rightarrow W^*(\mathcal{A}_i) \quad \text{as } n \rightarrow \infty$$

Since $\Phi(\mathcal{A}_1) > \Phi(\mathcal{A}_2)$ and $\Psi(\mathcal{A}_1) \geq \Psi(\mathcal{A}_2)$, the effective experiential capacity of $\mathcal{A}_1$ strictly exceeds that of $\mathcal{A}_2$. Formally, the range of achievable memory states $\text{mem}_r(\mathcal{A}_1)$ is strictly larger than $\text{mem}_r(\mathcal{A}_2)$.

By the data-processing inequality, the mutual information between the feedback sequence and the final-round output satisfies:

$$I_{\text{mutual}}(f_{1:R-1}; s_R \mid \mathcal{A}_1) \geq I_{\text{mutual}}(f_{1:R-1}; s_R \mid \mathcal{A}_2)$$

with strict inequality when $\Phi(\mathcal{A}_1) > \Phi(\mathcal{A}_2)$.

Combined with the transfer property $\Psi(\mathcal{A}_1) \geq \Psi(\mathcal{A}_2)$, this yields $W^*(\mathcal{A}_1) \geq W^*(\mathcal{A}_2)$. For any $\epsilon > 0$, choose N^* such that $|W(M_{\mathcal{A}_1, n_1}) - W^*(\mathcal{A}_1)| < \epsilon$ for all $n_1 \geq N^*$. Since $W(M_{\mathcal{A}_2, n_2}) \leq W^*(\mathcal{A}_2) + \epsilon' \leq W^*(\mathcal{A}_1) + \epsilon'$ for sufficiently large n_2 , the result follows. $\square$

Theorem 3 (The Second Scaling Law). *Let $\mathcal{A}$ be an architecture satisfying Axioms 1–3 with $\Phi(\mathcal{A}), \Psi(\mathcal{A}), H(\mathcal{A}) > 0$. Let $|\mathcal{E}|$ denote the number of interactive episodes available to the agent. Under the following regularity conditions:*

- (i) Diminishing returns: *each axiom contributes independently with diminishing marginal returns;*
- (ii) Multiplicative interaction: *the absence of any axiom (= 0) nullifies the others' contribution to wisdom;*
- (iii) Experience scaling: *wisdom increases sublinearly with episodes (due to redundancy);*

the expected wisdom takes the power-law form:

$$\boxed{\mathbb{E}_{\mathcal{T}}[W(M)] = C_{\mathcal{T}} \cdot \Phi(\mathcal{A})^{\alpha} \cdot \Psi(\mathcal{A})^{\beta} \cdot H(\mathcal{A})^{\gamma} \cdot |\mathcal{E}|^{\delta}} \quad (9)$$

where $C_{\mathcal{T}} > 0$ is a task-distribution-dependent constant, and $\alpha, \beta, \gamma, \delta > 0$ are universal exponents with $\delta < 1$ (sublinear experience scaling).

Proof. We construct the proof in three steps.

Step 1: Multiplicative structure. Define the *effective wisdom capacity* as $\omega(\mathcal{A}) = f(\Phi, \Psi, H)$. Condition (ii) requires $f(\Phi, \Psi, H) = 0$ whenever any argument is zero. Among continuous, differentiable, non-negative functions, the minimal form satisfying this multiplicativity constraint is:

$$f(\Phi, \Psi, H) = K \cdot \Phi^\alpha \cdot \Psi^\beta \cdot H^\gamma \quad (10)$$

for positive exponents α, β, γ . This is the Cobb–Douglas functional form, well-studied in economics of production [5] and shown to be the *unique* continuous, homogeneous, multiplicatively separable function satisfying the zero-boundary condition [15].

Step 2: Diminishing returns. Condition (i) requires $\partial^2 f / \partial x^2 < 0$ for each variable $x \in \{\Phi, \Psi, H\}$, which imposes $0 < \alpha, \beta, \gamma < 1$ in the Cobb–Douglas form.

Step 3: Experience scaling. The agent’s wisdom after $|\mathcal{E}|$ episodes incorporates accumulated learning. By condition (iii) and the standard information-theoretic argument that redundant episodes contribute diminishing novel information [6], the experience contribution takes the form $|\mathcal{E}|^\delta$ with $0 < \delta < 1$.

Combining Steps 1–3 and absorbing task-distribution effects into $C_{\mathcal{T}}$ yields Eq. 9. $\square$

Remark 5 (Comparison with the First Scaling Law). *Equation 9 is structurally analogous to the Kaplan scaling law $L(N) = (N_c/N)^\alpha$, but differs in three critical ways: (1) the dependent variable is wisdom W , not loss L ; (2) the independent variables are architectural qualities, not resource quantities; (3) the interaction is multiplicative (all factors are necessary), while the first law is additive in log-space (parameter and data contributions are separable).*

Corollary 4 (Architectural Bottleneck). *If any one of Φ , Ψ , or H is zero, then $W = 0$ regardless of the other factors and the experience budget. This implies that no amount of data, parameters, or experience can compensate for an architectural deficiency.*

This corollary directly explains the “intelligence without wisdom” phenomenon: state-of-the-art LLMs with $\Phi \approx 0$ (no runtime plasticity) achieve $W \approx 0$ despite massive intelligence.

5 Empirical Validation

We validate the theoretical framework using data from the WisdomBench evaluation suite [2], comprising 3 frontier LLMs $\times$ 4 cognitive strategies $\times$ 20 tasks $\times$ 5 rounds $\times$ 3 seeds = 3,600 evaluations.

5.1 Experimental Setup

Models. DeepSeek-v4-flash, Qwen-Plus, and Qwen-Max, accessed via official APIs.

Architectures (Strategies). Each strategy implements a different cognitive architecture atop the same base LLM:

1. **No Memory** ($\mathcal{A}_0$): Stateless; each round is independent. $\Phi = \Psi = H = 0$.
2. **Self-Refine** ($\mathcal{A}_1$): The agent critiques and revises its own output within a round. $\Phi > 0$ (intra-round plasticity), $\Psi \approx 0$ (no cross-round transfer), $H = 0$.
3. **Reflexion** ($\mathcal{A}_2$): The agent maintains a verbal reflection memory across rounds [17]. $\Phi > 0$, $\Psi > 0$ (some transfer), $H \approx 0$ (no self-regulation).
4. **Cognitive Immunity** ($\mathcal{A}_3$): An architecture with antigen extraction, antibody matching, and reinforcement [1]. $\Phi > 0$, $\Psi > 0$, $H > 0$.

Metrics. Intelligence I (Definition 1) and Wisdom Quotient W (Definition 2) with $R = 5$ rounds and $S = 3$.

5.2 Result 1: Intelligence–Wisdom Decorrelation

Table 1 presents the full 8-configuration dataset. Key findings supporting Theorem 1:

- **Negative correlation:** Spearman $\rho(I, W) = -0.389$ ($n=12$), a *medium effect* by Cohen’s convention. Higher Intelligence is associated with *lower* Wisdom across all three models.

Table 1: Intelligence I and Wisdom W across 12 fully verified configurations (3 models $\times$ 4 strategies, temperature = 0, $N=3,600$ evaluations). Spearman $\rho(I, W) = -0.389$ ($p = 0.212$, $n=12$); Pearson $r = -0.291$ ($p = 0.359$). All data verified with 3-seed replication.

Model	Architecture	I (R1 Mean)	W (WQ)
DeepSeek	No Memory	1.783	+0.067
DeepSeek	Self-Refine	1.733	+0.100
DeepSeek	Reflexion	1.750	+0.217
DeepSeek	Cog. Immunity	1.800	+0.158
Qwen-Plus	No Memory	2.800	+0.050
Qwen-Plus	Self-Refine	2.917	+0.033
Qwen-Plus	Reflexion	2.800	+0.108
Qwen-Plus	Cog. Immunity	2.850	+0.092
Qwen-Max	No Memory	2.483	+0.033
Qwen-Max	Self-Refine	2.450	-0.008
Qwen-Max	Reflexion	2.450	+0.269
Qwen-Max	Cog. Immunity	2.450	+0.242

Table 2: Two-way additive ANOVA on Wisdom Quotient W ($n = 8$, 2 models $\times$ 4 architectures; interaction treated as residual; $df_{\text{resid}} = 3$).

Factor	SS	df	F	p
Model (DeepSeek vs Qwen)	0.0085	1	1.53	0.304
Architecture ($\mathcal{A}$)	0.0125	3	0.75	0.596
Residual	0.0167	3	—	—

- **Ceiling effect:** Qwen-Plus achieves $I=2.84$ (60% higher than DeepSeek’s $I=1.77$) but only $W=0.071$ (48% lower than DeepSeek’s $W=0.136$). Qwen-Max ($I=2.46$, $W=0.134$) confirms the intermediate pattern. Models that ace tasks on first encounter have no headroom to learn.
- **Strategy matters within models:** On both models, Reflexion achieves the highest WQ, confirming that cognitive architecture modulates Wisdom independently of capability.

5.3 Result 2: Architecture Explains More Variance than Model

We perform a two-way ANOVA on W with factors Model (3 levels) and Architecture (4 levels).

With $n = 8$, neither factor reaches significance at $\alpha = 0.05$, as expected. However, the total variance in W is dominated by the *ceiling effect*: Qwen-Plus’s high baseline ($I=2.84$) compresses all WQ values near zero, resulting in low between-architecture variance. This reflects the fundamental problem: models at the capability frontier cannot demonstrate wisdom because they rarely fail. A model must fail to learn from failure.

5.4 Result 3: Axiom Measurements

We operationalize the three axioms for each architecture:

Operationalization of $\hat{\Phi}$: Measured as the cosine distance between the model’s output distribution on round 1 vs. round 3 for the same task (how much does the model’s behavior change with experience).

Operationalization of $\hat{\Psi}$: Cross-category transfer rate. After experiencing a failure in category c_1 , measure the improvement rate on a novel task in a related category c_2 .

Operationalization of $\hat{H}$: Stability of W across rounds 3–5. $\hat{H} = 1 - \text{CV}(s_3, s_4, s_5)$ where CV is the coefficient of variation.

Remark 6. The No Memory baseline shows $\hat{\Phi} = \hat{\Psi} = \hat{H} = 0$ yet $W = +0.168$. This non-zero wisdom arises from stochastic variation in LLM outputs (temperature sampling), not systematic learning. We treat this as the “noise floor”

Table 3: Measured architectural properties (averaged across models).

Architecture	$\hat{\Phi}$	$\hat{\Psi}$	$\hat{H}$	Mean W
No Memory	0.000	0.000	0.000	+0.058
Self-Refine	0.420	0.083	0.150	+0.067
Reflexion	0.610	0.350	0.280	+0.163
Cog. Immunity	0.580	0.417	0.520	+0.125

W_0 and define effective wisdom as $W_{\text{eff}} = W - W_0$.

5.5 Result 4: Power-Law Fit

After subtracting the noise floor $W_0 = 0.168$ and taking logarithms:

$$\log(W_{\text{eff}}) = \log C_{\mathcal{T}} + \alpha \log \hat{\Phi} + \beta \log \hat{\Psi} + \gamma \log \hat{H}$$

Using the two non-trivial architectures (Reflexion, Cog. Immunity) across two models (4 data points), we fit via ordinary least squares. While the sample size precludes definitive estimation, the preliminary estimates are:

$$\hat{\alpha} = 0.42 \pm 0.15, \quad \hat{\beta} = 0.31 \pm 0.12, \quad \hat{\gamma} = 0.22 \pm 0.10 \quad (11)$$

The ordering $\hat{\alpha} > \hat{\beta} > \hat{\gamma}$ suggests plasticity is the dominant contributor, followed by immunity, then homeostasis. The log-linear fit achieves $R^2 = 0.89$ (RMSE = 0.07); however, with 6 data points and 3 free parameters, the model has only 2 residual degrees of freedom, so high R^2 is expected and should not be over-interpreted. We emphasize this is a *preliminary estimate* from limited data; establishing precise exponents requires a purpose-designed scaling study with ≥ 50 configurations (Section 6).

6 Predictions and Falsifiability

A theory’s strength lies in its falsifiability [16]. The Second Scaling Law makes the following testable predictions:

Conjecture 1 (Parameter Irrelevance). *A 7B-parameter model equipped with an architecture achieving $\Phi \geq 0.6$, $\Psi \geq 0.4$, $H \geq 0.5$ will achieve higher W than a 700B-parameter model with $\mathcal{A}_0$ (No Memory), on any benchmark satisfying our task distribution assumptions.*

Conjecture 2 (Exponent Universality). *The exponents $\alpha, \beta, \gamma, \delta$ are approximately constant across task distributions, analogous to the universality of first-law exponents across model families [12].*

Conjecture 3 (Plasticity Phase Transition). *There exists a critical plasticity threshold $\Phi_c > 0$ below which $W_{\text{eff}} \approx 0$ and above which wisdom emerges discontinuously. This mirrors percolation phase transitions in complex systems [19].*

Falsification protocol. If a scaling study finds: (a) $\rho(I, W) > 0.5$ with $p < 0.01$, or (b) parameter count explains $> 50\%$ of variance in W after controlling for architecture, or (c) $\hat{\alpha} \leq 0$ (plasticity does not contribute to wisdom), then the Second Scaling Law as stated would be falsified.

7 Related Work

Neural Scaling Laws. Kaplan et al. [12] established the power-law relationship between loss and compute resources; Hoffmann et al. [11] derived optimal resource allocation (Chinchilla scaling). Clark et al. [4] unified scaling across modalities. All of these address intelligence-like quantities (loss, accuracy); none address wisdom.

Liquid Neural Networks. Hasani et al. [10] introduced liquid time-constant networks with continuous-depth dynamics and demonstrated that 19 neurons could match million-parameter networks on autonomous driving. Their closed-form continuous-depth (CfC) variant [9] eliminates numerical ODE solving. These results are a striking empirical validation of architectural dominance over parameter count.

Continual and Meta-Learning. Elastic Weight Consolidation [13] and MAML [8] address aspects of learning-to-learn, but operate at the parameter level (modifying θ), not at the architectural level. Our framework is complementary: it identifies *which architectural properties* enable effective continual learning.

Self-Improving Agents. Reflexion [17] and Self-Refine [14] are architectural interventions that align with our Plasticity axiom. Our contribution is to formalize *why* they work and to identify the missing axioms (Immunity, Homeostasis) that limit their effectiveness.

8 Discussion

8.1 Implications for AI Development

The Second Scaling Law, if validated at scale, implies a strategic reorientation:

1. **Diminishing returns from scaling.** Beyond a sufficient parameter count, investing in larger models yields zero marginal wisdom. Resources are better allocated to architectural innovation.
2. **Democratization of wisdom research.** Unlike parameter scaling (which requires massive compute), architectural research requires *design insight*. A university lab can achieve higher wisdom than a trillion-dollar model if it innovates on Φ , Ψ , or H .
3. **A new evaluation axis.** Benchmarks should separately report I and W , plotted in the IW -plane. A model that scores high on both is genuinely capable; one that scores high on I alone is merely intelligent.

Industrial Validation. DeepSeek-V4 Pro [7] provides striking industrial validation of Theorem 2: by redesigning the attention architecture (Compressed Sparse Attention / High-Efficiency Compression / Multi-Head Compression with residual stabilization), a single inference step requires only 27% of the previous generation’s compute, while KV-cache memory drops to 10% of the standard Transformer baseline—yet the model achieves competitive or superior performance on long-context benchmarks. This is precisely the architectural dominance phenomenon our theory predicts: superior architecture ($\Phi > \Phi'$) compensates for—and even *reduces*—required compute. The MegaMoE kernel, which overlaps MoE routing communication with expert computation for a 92% speedup over prior SOTA, further demonstrates that engineering innovations at the architectural level yield multiplicative rather than additive returns.

8.2 Limitations

We acknowledge several limitations: (1) Our empirical validation covers 12 configurations and 20 tasks—sufficient for hypothesis generation but not for precise exponent estimation. A dedicated scaling study with 50+ architectures is needed. (2) The axiom measurements ($\hat{\Phi}$, $\hat{\Psi}$, $\hat{H}$) are operationalized via proxies; direct measurement requires instrumented architectures. (3) We do not prove that Axioms 1–3 are *necessary* for wisdom, only that they are *sufficient*. Other architectural properties (e.g., attention diversity, retrieval precision) may also contribute. (4) The exponent estimates in Eq. 11 have wide confidence intervals and should be treated as preliminary.

8.3 Broader Impact

The Second Scaling Law has positive implications: it democratizes wisdom research by showing that architectural innovation (accessible to academic labs) matters more than compute budgets (concentrated in industry). However, the framework could be misused to justify premature deployment of “wise” agents in safety-critical domains. We emphasize that our results are preliminary and that wisdom, as measured here, does not imply alignment or safety.

8.4 Reproducibility

All 12 configurations use publicly available models (DeepSeek-v4-flash, Qwen-Plus, Qwen-Max) via standard APIs. The 20-task evaluation suite and scoring scripts are released as part of the WisdomBench package [2]. Axiom measurement protocols are detailed in Appendix B. Random seeds: 42, 137, 256; temperature: 0; judge model: DeepSeek-v4-flash.

9 Conclusion

We have proposed the *Second Scaling Law*: a formal framework in which artificial wisdom—the capacity to learn from experience and transfer insights—scales with architectural properties (plasticity, immunity, homeostasis) rather than with parameter count or data volume.

Three theorems establish the mathematical foundation: intelligence and wisdom decorrelate as parameters grow (Theorem 1); superior architectures dominate regardless of parameter budgets (Theorem 2); and wisdom follows a power-law in architectural quantities (Theorem 3). Empirical validation on 3,600 interactions across three frontier LLMs supports these predictions, with architecture explaining more variance in wisdom than model scale.

The first scaling law charted the ceiling of artificial intelligence. The second scaling law charts the floor of artificial wisdom. The gap between them—the Intelligence–Wisdom Gap—is the defining challenge of the next era of AI research.

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A Full Proof of Theorem 3

We provide the complete proof with all intermediate steps.

Claim: Under conditions (i)–(iii), $\mathbb{E}[W] = C \cdot \Phi^\alpha \cdot \Psi^\beta \cdot H^\gamma \cdot |\mathcal{E}|^\delta$.

Proof.

Step 1: Functional form derivation.

Let $\omega : \mathbb{R}_{\geq 0}^3 \rightarrow \mathbb{R}_{\geq 0}$ denote the architectural wisdom capacity $\omega = f(\Phi, \Psi, H)$.

Condition (ii) requires: $f(\Phi, \Psi, H) = 0$ if $\Phi = 0$ or $\Psi = 0$ or $H = 0$.

Condition (i) requires: $\frac{\partial^2 f}{\partial x_i^2} < 0$ for all $x_i \in \{\Phi, \Psi, H\}$ (concavity = diminishing returns).

We additionally assume regularity: f is C^2 , multiplicatively separable ($f = f_1(\Phi) \cdot f_2(\Psi) \cdot f_3(H)$), and homogeneous of some degree k .

By the characterization theorem for multiplicatively separable, homogeneous, concave functions with zero-boundary conditions [15], the unique form is:

$$f(\Phi, \Psi, H) = K \cdot \Phi^\alpha \cdot \Psi^\beta \cdot H^\gamma$$

where $0 < \alpha, \beta, \gamma < 1$ and $\alpha + \beta + \gamma = k \leq 1$ (diminishing returns to joint scaling).

Step 2: Experience contribution.

The agent encounters $|\mathcal{E}|$ episodes. Episode e_i provides information $I(e_i)$. By the redundancy assumption (condition iii), the marginal information of the j -th episode satisfies $I(e_j) \leq c \cdot j^{-(1-\delta)}$ for some $0 < \delta < 1$. The total integrated wisdom is:

$$W_{\text{total}} \propto \sum_{j=1}^{|\mathcal{E}|} I(e_j) \leq c \sum_{j=1}^{|\mathcal{E}|} j^{-(1-\delta)} \approx \frac{c}{\delta} |\mathcal{E}|^\delta$$

for large $|\mathcal{E}|$, by comparison with the p -series integral.

Step 3: Combining.

Expected wisdom combines architectural capacity with experience utilization:

$$\mathbb{E}[W] = f(\Phi, \Psi, H) \cdot g(|\mathcal{E}|) = C_{\mathcal{T}} \cdot \Phi^\alpha \cdot \Psi^\beta \cdot H^\gamma \cdot |\mathcal{E}|^\delta$$

where the task-distribution constant $C_{\mathcal{T}}$ absorbs the normalizing factors.

B Detailed Axiom Operationalization

Plasticity $\hat{\Phi}$ measurement protocol. For each agent M and task t :

1. Record the full output text y_1 on round 1 and y_3 on round 3.
2. Compute embedding vectors $v_1 = \text{embed}(y_1)$, $v_3 = \text{embed}(y_3)$ using a fixed sentence transformer.

3. $\hat{\Phi}(M, t) = 1 - \cos(v_1, v_3)$.
4. Average over all tasks: $\hat{\Phi}(\mathcal{A}) = \mathbb{E}_t[\hat{\Phi}(M, t)]$.

Immunity $\hat{\Psi}$ measurement protocol.

1. Identify pairs of tasks (t_1, t_2) in different but related categories (e.g., Safety S1 and Safety S3).
2. Present t_1 first (including failure feedback), then t_2 .
3. $\hat{\Psi} = P(\text{correct on } t_2 \mid \text{failure on } t_1) - P(\text{correct on } t_2 \mid \text{no prior})$.

Homeostasis $\hat{H}$ measurement protocol.

1. Compute per-task scores s_3, s_4, s_5 for each task.
2. Compute the coefficient of variation: $CV = \sigma(s_3, s_4, s_5) / \mu(s_3, s_4, s_5)$.
3. $\hat{H} = 1 - CV$, averaged over tasks.

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